

KLH (DEEMED TO BE UNIVERSITY)

Database Systems Engineering and Distributed Backend Development (25CS1302E)

Food Donation & NGO Connect Platform

A smart, location-aware rescue network connecting donors, volunteers and NGOs

REVIEW 2 • Sec No-10 • Team No-18

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Problem Statement

Restaurants, households and events generate surplus edible food every day, but getting it to the people who need it still depends on slow, fragmented coordination.

1

Scattered channels

Phone calls and WhatsApp groups are unorganised and easily missed.

2

No location context

Donors cannot see which NGO is nearby, and NGOs cannot see what is close to them.

3

Food expires

Cooked meals have short windows, so delays in finding an NGO mean waste.

4

No tracking

Nothing records who collected the food or whether it was delivered.

Result: good food expires, NGOs go without, and the same shortages repeat across the city.

Objectives

1

Build a centralized relational platform to list and track food donations

2

Enable location-aware matching of donors to nearby NGOs

3

Automatically rank matches by distance, quantity and expiry window

4

Verify donors and NGOs before food is listed or claimed

5

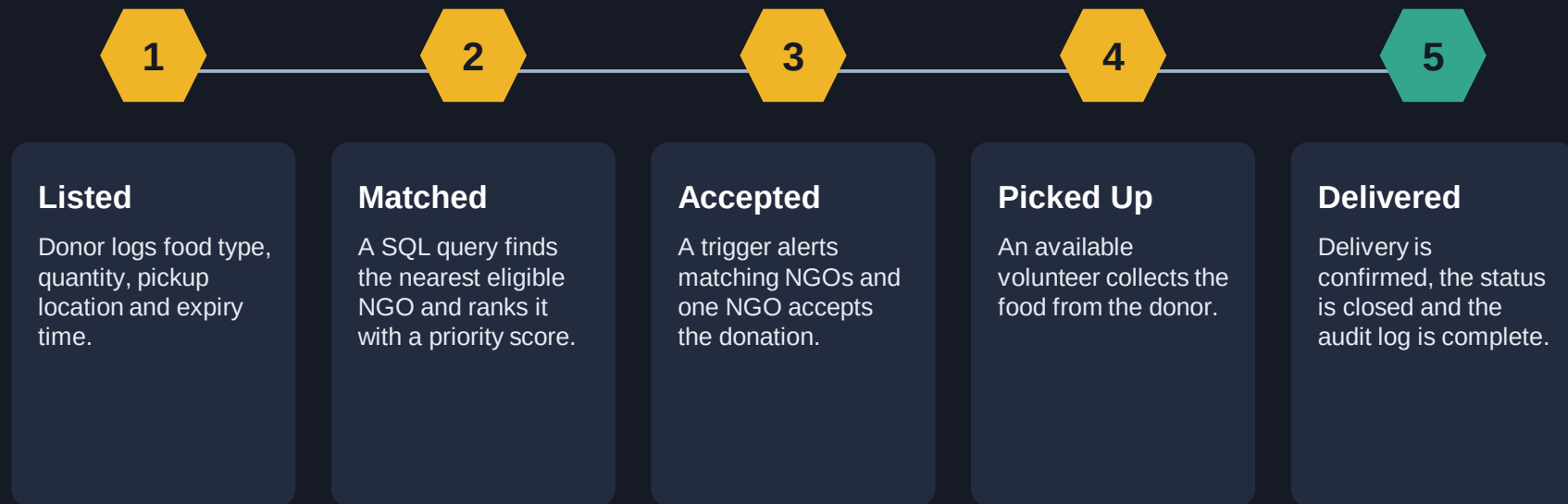
Track every donation through a clear pickup-to-delivery lifecycle

6

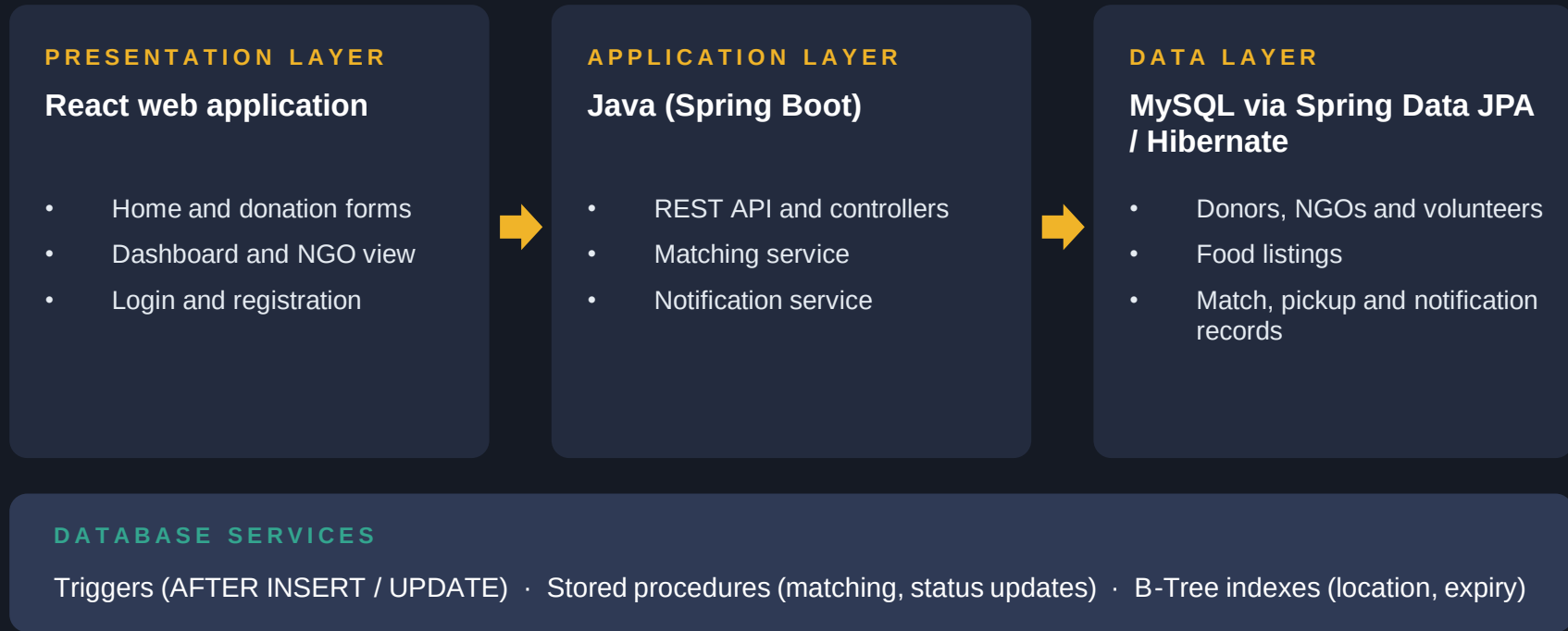
Give administrators tools to review, verify and monitor activity

Proposed System

The platform gives every donation a rescue trail, from the moment it is listed to the moment it reaches an NGO.



System Architecture



Modules / Functionalities



Auth & Profiles

Registration, login and role-based access for donors, NGOs, volunteers and admins



Donation Listing

Capture food type, quantity, pickup location and expiry time



Matching Engine

SQL joins and proximity filters rank NGOs by distance, quantity and expiry



Alerts & Triggers

Triggers update status and notify NGOs when a matching donation is added



Pickup & Delivery

Assign available volunteers and track status until delivery



Verification

Admins verify donors and NGOs before they can list or claim food



Admin Panel

Review listings, resolve disputes and monitor all activity



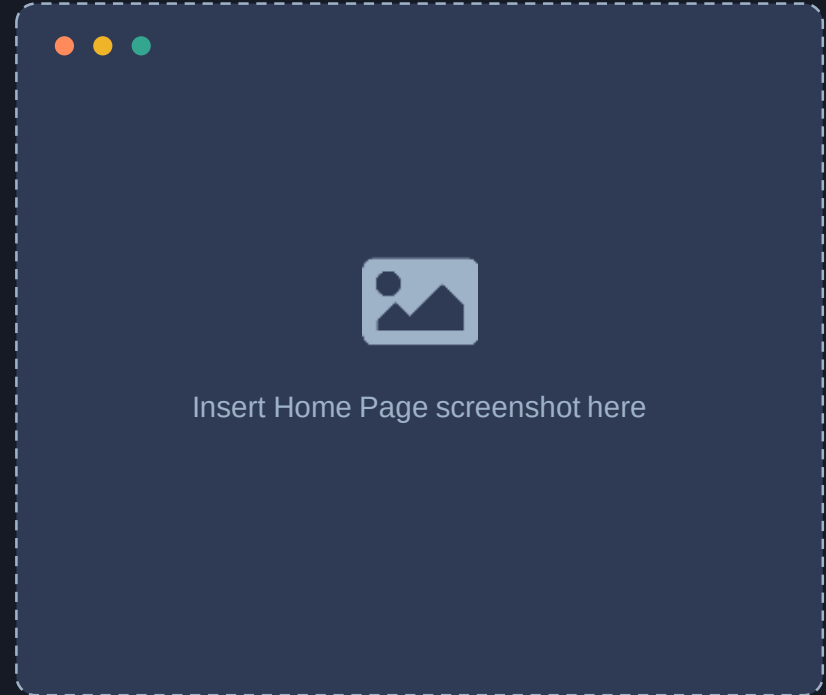
Reports & Insights

Donation history, audit trail and wastage-reduction trends

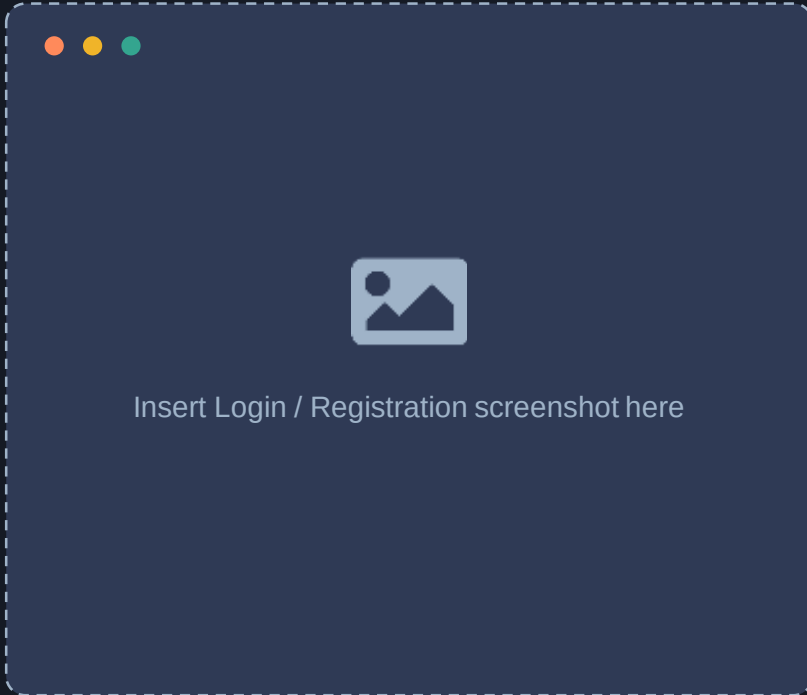
Frontend Design: Home Page

DESIGN HIGHLIGHTS

- Clear navigation to List Donation, Find NGOs and My Donations
- Summary cards for active, matched, picked up and delivered donations
- Recently listed donations shown as scannable cards with expiry time
- Consistent header and footer used across every page



Frontend Design: Login



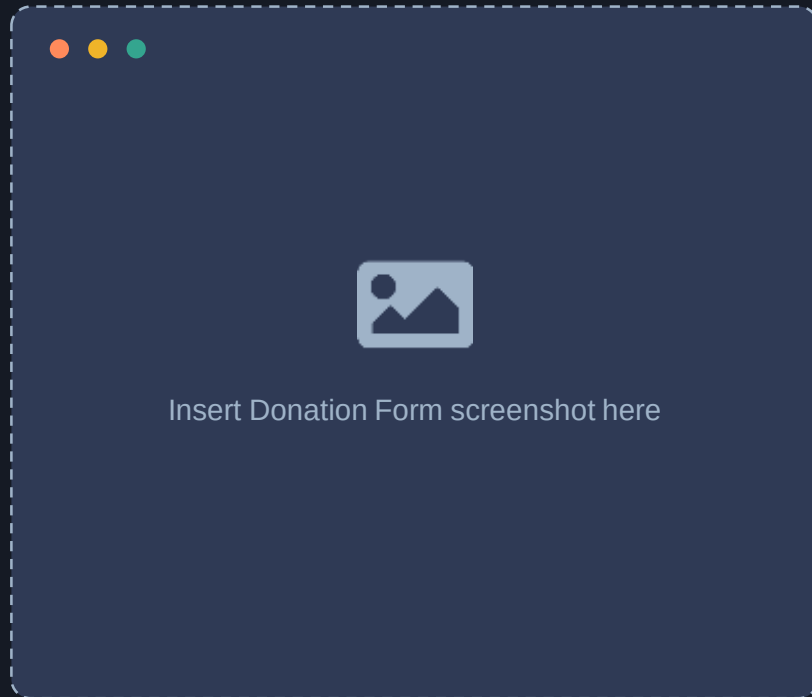
DESIGN HIGHLIGHTS

- Separate sign-up for donors, NGOs and volunteers
- Form validation for email, phone and password strength
- Session handling for secure, role-based access
- Clean, distraction-free layout for fast entry

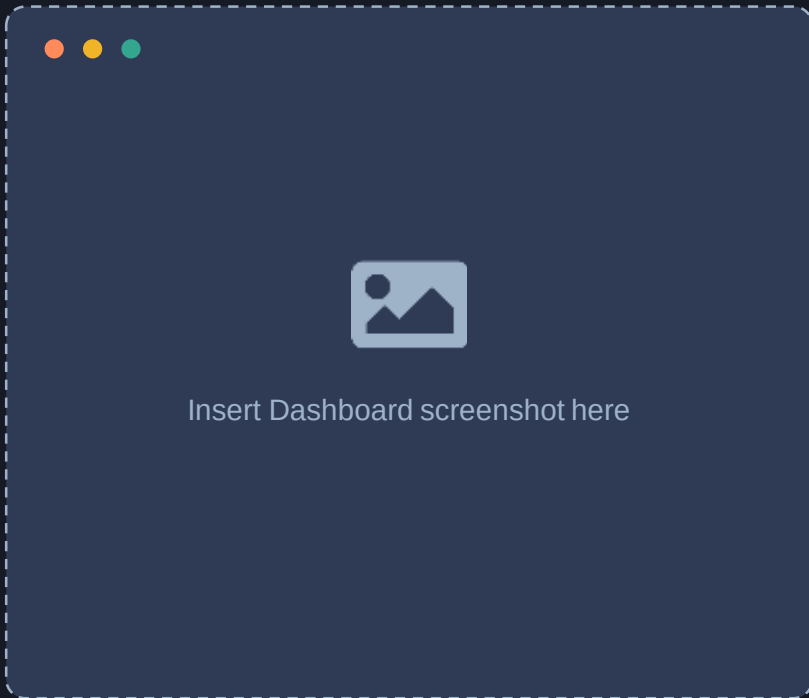
Frontend Design: Donation Form

DESIGN HIGHLIGHTS

- Fields for food type, quantity, pickup address and expiry time
- Preferred pickup window to help volunteers plan collection
- Optional notes on packaging or handling
- Inline validation before submission



Frontend Design: Dashboard



DESIGN HIGHLIGHTS

-  My Donations view with live lifecycle status
-  Match alerts for NGOs surfaced at the top
-  Filter by food type, distance or status
-  Admin view adds verification and monitoring controls

Database Design: ER Diagram



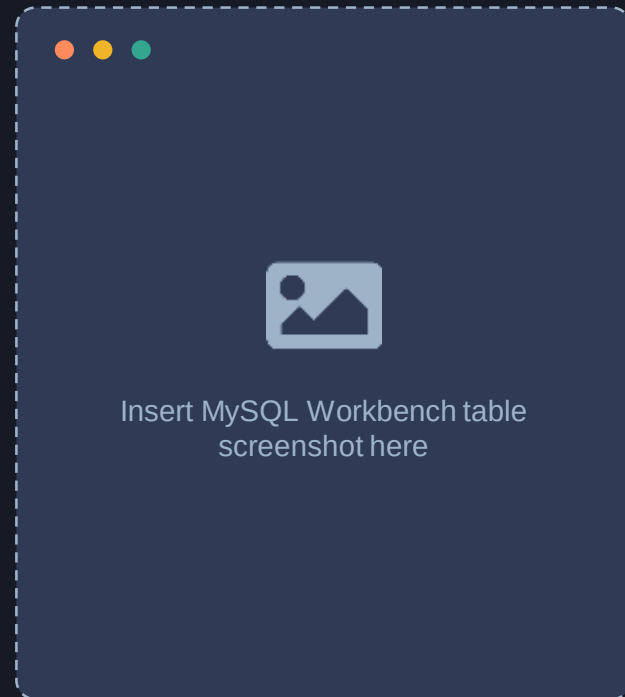
Database Design: Schema & Samples

FOOD_LISTING: KEY COLUMNS

Column	Type	Notes
listing_id	BIGINT (PK)	Auto-increment
donor_id	BIGINT (FK)	References DONOR
food_type	VARCHAR(50)	e.g. Cooked meals, Bakery
quantity_kg	DECIMAL(6,2)	Approximate quantity
expiry_time	DATETIME	Used in the matching window
status	VARCHAR(30)	Lifecycle stage

SAMPLE RECORDS (ILLUSTRATIVE)

ID	Food type	Qty (kg)	Status
101	Cooked meals	45	Matched
102	Bakery items	12	Listed
103	Packed snacks	20	Delivered



Current Progress



COMPLETED

-  Abstract finalised and submitted
-  Requirements, objectives and scope defined
-  System architecture designed
-  ER diagram and database schema drafted
-  Tech stack finalised (Spring Boot, MySQL, React)



IN PROGRESS / PENDING

-  Frontend page implementation (Home, Forms, Dashboard)
-  Backend REST APIs and Spring Boot setup
-  SQL matching procedure and triggers
-  Pickup tracking and notification flow
-  Admin panel and verification workflow

Challenges & Future Work

CHALLENGES / ISSUES FACED

-  Balancing distance, quantity and expiry so the best NGO is ranked first
-  Keeping match and status updates consistent when donations arrive together
-  Coordinating schema design across three team members

FUTURE WORK

-  Complete backend APIs and connect to the MySQL database
-  Implement matching procedures and triggers
-  Build and refine all frontend pages with real data
-  Test the matching logic against realistic sample data

CONCLUSION

Turning surplus food into a network

-  Centralizes donor, NGO and volunteer records in one relational database
-  Uses SQL matching and triggers to speed up food rescue
-  Tracks every pickup with a full audit trail
-  Gives administrators visibility, verification and control



Thank You

Team 18

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